

Abstract

This study examines whether tourism contributes to economic growth across Indian states or if its role is overstated. Using panel data for 15 states from 2011 to 2023, the analysis evaluates the relationship between tourism and economic performance within a broader economic framework. Economic growth is measured by Gross State Domestic Product (GSDP), while tourism is captured through Domestic Tourist Visits and Foreign Tourist Arrivals. Population and government capital outlay are included as control variables to account for internal economic capacity and public investment. The study follows a structured econometric approach, including descriptive analysis, correlation testing, and diagnostic checks for non-stationarity, heteroskedasticity, and cross-sectional dependence. Based on these tests, a Fixed Effects model with Driscoll-Kraay standard errors is employed to obtain robust estimates. The results show that domestic tourism has a positive and consistent impact on economic growth. In contrast, foreign tourism does not exhibit a stable or significant contribution. More importantly, population and government investment have a stronger influence on economic performance than tourism. The findings indicate that tourism is not a primary driver of growth but a supplementary sector whose impact depends on regional conditions and supporting infrastructure. This highlights the need for targeted, state-specific tourism policies rather than generalized strategies.

Keywords: Tourism Development, Economic Growth, GSDP, Panel Data, Fixed Effects Model, Driscoll-Kraay, Domestic Tourism, Foreign Tourism, India

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Over the past few decades, the global economy has shifted from agriculture and industry toward the service sector with tourism emerging as a contributor. Tourism has evolved from a leisure activity into an economic sector that supports growth particularly in developing economies. Tourism generates both direct and indirect economic benefits. Tourist spending supports accommodation, transport, local businesses and informal sectors, creating a multiplier effect within the economy. It also promotes infrastructure development, including roads, airports, and public services. In developing regions tourism can support employment generation, income creation, and regional development by directing investment into less developed areas. It also contributes to foreign exchange earnings and attracts public and private investment. However tourism does not automatically lead to economic growth. The relationship depends on underlying economic conditions, infrastructure, and policy effectiveness. Therefore, empirical analysis is necessary to evaluate the actual impact of tourism on economic performance and to guide effective policy decisions.

1.2 Tourism and Economic Growth

The relationship between tourism and economic growth is explained by the Tourism-Led Growth Hypothesis (TLGH), which argues that tourism can directly contribute to economic expansion rather than merely result from it. Increased tourist inflow raises demand for accommodation, transport, food services, and related sectors, generating a multiplier effect across the economy. Early studies relied on time series analysis focused on individual countries, which failed to capture regional differences. Recent research uses panel data analysis to examine multiple regions over time, providing a more comprehensive understanding of the tourism growth relationship. Common methods include Pooled OLS, Fixed Effects, and Random Effects models. However, panel analysis involves key econometric challenges. Unobserved heterogeneity arises from region-specific factors such as infrastructure and policy differences. Heteroskedasticity leads to unequal variance across regions, and cross-sectional dependence reflects interlinkages between states through trade and economic interactions. These issues indicate that the tourism growth relationship is complex and requires robust econometric techniques to produce reliable results.

1.3 Importance of Tourism in India

India's tourism sector plays a significant role in economic activity due to its geographical, cultural, and heritage diversity. The country attracts both domestic and international tourists through varied destinations such as beaches, heritage sites, and religious centers. Tourism contributes to employment generation, supports local livelihoods, and serves as a source of foreign exchange earnings. As part of the service sector, it has become an important component of India's economic structure. However, the impact of tourism varies across states. Due to India's federal structure, states differ in tourism development, infrastructure, and policy implementation. States like Kerala, Rajasthan, and Goa have effectively utilized tourism for economic growth through targeted strategies and strong branding. In contrast, other states have not achieved similar outcomes due to limitations in infrastructure, connectivity, and promotion. This uneven development suggests that the impact of tourism on economic growth depends on regional conditions rather than being uniform across all states. Therefore, analysing tourism across states is essential to understand its actual contribution to economic growth in India.

1.4 Statement of the Problem

India has experienced consistent growth in domestic and international tourism, leading to the common assumption that tourism promotes economic development. However, it remains unclear whether tourism contributes to economic growth uniformly across states or whether its impact depends on regional conditions. The extent and nature of its influence on Gross State Domestic Product (GSDP), particularly in the long run, are not well established. Observing simultaneous growth in tourism and the economy does not confirm a causal relationship. Existing studies often rely on simplified methods that fail to capture the interconnected nature of Indian states. Economic developments in one state can influence others through infrastructure, policy, and investment linkages, creating cross-sectional dependence. Additionally, unobserved heterogeneity and heteroskedasticity across states complicate reliable estimation. These limitations highlight the need for a more robust analytical framework. This study addresses the gap by applying panel data techniques that account for cross-sectional dependence and heteroskedasticity. Using unit root and cointegration tests along with a Fixed Effects model and Driscoll-Kraay standard errors, the study aims to provide more reliable evidence on the relationship between tourism and economic growth across Indian states.

1.5 Research Questions

Considering the aforementioned objectives this study attempts to provide answers to the following research questions:

1. Is there any cointegrating relationship between tourism development indicators and Gross State Domestic Product (GSDP) in the chosen states of India in the long run?
2. What is the precise impact of Domestic Tourist Arrivals (DTA) vis à vis Foreign Tourist Arrivals (FTA) on economic growth?
3. What is the impact of cross section dependence, together with panel heteroscedasticity, on the aforementioned relationship and how can Driscoll Kraay standard errors help address these issues?

1.6 Objectives of the Study

The objective of the study is to empirically investigate the complex relationships between tourism development and economic growth for a panel of Indian states. The specific objectives of the study are as follows:

1. To empirically test for the existence of a Long-term relationship between tourism development (measured by domestic and foreign tourist arrivals) and economic growth (measured by GSDP) for 15 major Indian states over the study period.
2. To investigate the relative importance of tourism indicators and state infrastructure on economic output.
3. To empirically investigate the relative impact of domestic tourism and foreign tourism on economic growth in Indian states, while fully addressing the issue of cross section dependence and state specific heterogeneity.

1.7 Data Source and Methodology

This study uses secondary data obtained from official and reliable government sources to examine the relationship between tourism and economic growth across Indian states. All variables are selected to ensure consistency, comparability, and credibility. Economic growth is measured using Gross State Domestic Product (GSDP), sourced from the Ministry of Statistics and Programme Implementation. Tourism activity is captured through Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA), obtained from the Ministry of Tourism. Population data is collected from the Census of India and supplemented with state statistical abstracts. Capital Outlay (CAP), representing government investment, is sourced from the Reserve Bank of India. The dataset is structured as a balanced panel covering 15 Indian states over the period 2011-2023. This allows analysis of both time variation within states and differences across states. The study follows a systematic econometric approach. It begins with descriptive statistics and correlation analysis, followed by panel unit root tests to examine stationarity and cointegration tests to identify long-run relationships. For estimation,

Pooled OLS, Fixed Effects, and Random Effects models are initially considered. The Hausman test is used to select the appropriate model. Given the presence of cross-sectional dependence, heteroskedasticity, and autocorrelation, the final estimation employs a Fixed Effects model with Driscoll-Kraay standard errors to ensure robust and reliable results.

1.8 Scope of the Study

This study analyzes the impact of tourism on economic growth across 15 Indian states using panel data from 2011 to 2023. The selected period captures recent changes in tourism development, infrastructure, and policy. The study follows a structured econometric framework. It begins with descriptive statistics, correlation analysis, and multicollinearity checks. Stationarity is tested using the Im-Pesaran-Shin (IPS) unit root test, and long-run relationships are examined using the Kao cointegration test. Panel regression models, including Pooled OLS, Fixed Effects, and Random Effects, are estimated, with the Hausman test used for model selection. To address cross-sectional dependence and heteroskedasticity, the final model applies Fixed Effects with Driscoll-Kraay standard errors. Economic growth is measured by Gross State Domestic Product (GSDP). Tourism is represented by Domestic Tourist Arrivals and Foreign Tourist Arrivals, while capital outlay is included as a proxy for infrastructure. The study is limited to selected states, variables, and time period, focusing on providing reliable evidence on the tourism growth relationship in India.

1.9 Organization of the Project

Chapter 1 - Introduces the study by outlining the background, statement of the problem, objectives, scope, data sources, and methodology.

Chapter 2 - Reviews the existing literature on tourism and economic growth at both global and Indian levels, and identifies the research gap.

Chapter 3 - Presents the data and econometric methodology, including variable definitions, model specifications, and diagnostic tests.

Chapter 4 - Analyzes the empirical results, including descriptive statistics, regression outcomes, and diagnostic testing, and interprets the findings.

Chapter 5 - Summarizes the key findings, provides policy implications, and concludes the study.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

The relationship between tourism and economic growth has been widely studied, leading to the development of the Tourism-Led Growth Hypothesis (TLGH). However, existing evidence is mixed, with some studies supporting a positive impact while others find weak or conditional relationships. This makes it necessary to critically examine prior research and its methodological approaches. This chapter reviews empirical literature on tourism and economic growth to identify findings, patterns, and limitations. It begins with global studies, including cross-country and time series analyses, which provide broad insights but often overlook regional differences. The review then focuses on India, where state-level variations in tourism and economic development require more localized analysis. The literature also shows a shift from simple time series methods to panel data techniques, reflecting the need to capture both cross-sectional and time-based variations. Despite these advancements, many studies fail to address issues such as cross-sectional dependence, heterogeneity, and other econometric challenges. These gaps highlight the need for a more robust analytical framework, which forms the basis for the present study.

2.2 Review of Studies on the Global Tourism Economy

The study of tourism and economic growth has evolved through a combination of empirical evidence and methodological advancements. The literature reflects a gradual shift from simple time series analysis to more complex panel data approaches, alongside improvements in econometric techniques.

Balaguer and Cantavella-Jordá (2002), titled “Tourism as a Long-Run Economic Growth Factor: The Spanish Case,” examines the long-run relationship between tourism receipts and economic growth in Spain. Using cointegration and causality analysis, the study finds that tourism and economic growth are closely linked in the long run, with causality running from tourism to growth. This provides strong support for the Tourism-Led Growth Hypothesis (TLGH).

Durbarry (2004), in “Tourism and Economic Growth: The Case of Mauritius,” analyzes the role of tourism in a developing island economy. The study finds that tourism has a strong positive impact on economic growth, mainly through multiplier effects, highlighting its importance in small and tourism-dependent economies.

Oh (2005), in “The Contribution of Tourism Development to Economic Growth in the Korean Economy,” tests the direction of causality between tourism and growth. The findings indicate reverse causality, where economic growth drives tourism, suggesting that tourism is not always the primary source of growth.

Lee and Chang (2008) in “Tourism Development and Economic Growth: A Closer Look at Panels.” Using panel data across developed and developing countries, the study finds that tourism leads economic growth in developed economies, while a bidirectional relationship exists in developing countries, indicating mutual reinforcement.

Sequeira and Campos (2005), in “International Tourism and Economic Growth: A Panel Data Approach,” examines cross-country data and finds that tourism positively influences economic growth, with stronger effects observed in developing economies.

Narayan, Narayan and Prasad (2010), titled “Tourism and Economic Growth: A Panel Data Analysis for Pacific Island Countries,” focuses on island economies and finds strong support for TLGH, with tourism acting as a major driver of long-run economic growth.

Cortés-Jiménez and Pulina (2010), in “Inbound Tourism and Long-Run Economic Growth,” analyzes the role of inbound tourism and finds that it positively affects economic growth, although the impact varies across countries.

Katircioglu (2009), titled “Tourism, Trade and Growth: The Case of Cyprus,” examines the interaction between tourism, trade, and growth. The findings suggest that tourism contributes to economic growth, but its effectiveness depends on trade openness and economic structure.

Tang and Tan (2015), in “Does Tourism Effectively Stimulate Malaysia’s Economic Growth?” uses advanced econometric techniques and concludes that tourism supports growth, but its effectiveness depends on policy environment, infrastructure, and institutional quality.

Brida, Cortés-Jiménez and Pulina (2016), in “Has the Tourism-Led Growth Hypothesis Been Validated?” provides a comprehensive review of existing studies. The authors conclude that while most studies support TLGH, the results vary significantly depending on methodology, data, and country-specific conditions.

Econometric Foundations

Dickey and Fuller (1979), “Distribution of the Estimators for Autoregressive Time Series with a Unit Root,” addresses the problem of non-stationarity in time series data. Macroeconomic variables such as GDP and tourism often exhibit trends, leading to spurious regression results if used directly. To address this, the authors developed the Dickey - Fuller (DF) unit root test to determine whether a variable is stationary. The null hypothesis states that the series has a

unit root (non-stationary). This contribution forms the foundation of modern econometric analysis, as testing for stationarity is essential before any regression is performed.

Phillips and Perron (1988), in “Testing for a Unit Root in Time Series Regression,” identified limitations in the DF test, particularly its sensitivity to autocorrelation and heteroskedasticity. They developed the Phillips-Perron (PP) test, which adjusts test statistics to account for these issues without modifying the regression structure. The hypotheses remain the same as the DF test, but the PP approach provides more reliable results under real-world data conditions.

Im-Pesaran and Shin (2003), “Testing for Unit Roots in Heterogeneous Panels.” This study extends unit root testing to panel data by allowing heterogeneity across cross-sectional units. The IPS test combines individual Augmented Dickey-Fuller regressions and tests whether all panels are non-stationary against the alternative that some are stationary. This approach is more realistic for studies involving multiple regions, such as Indian states.

Kao (1999), in “Spurious Regression and Residual-Based Tests for Cointegration in Panel Data,” extends cointegration analysis to panel settings. The Kao test examines whether variables move together in the long run despite being individually non-stationary. The null hypothesis states no cointegration, while rejection indicates a valid long-run equilibrium relationship. This ensures that regression results are meaningful rather than spurious.

M. Hashem Pesaran (2004, 2007) introduced the Cross-Section Dependence (CD) test to detect interdependence across units. The test checks whether residuals are correlated across panels. Rejection of the null hypothesis indicates dependence, implying that standard estimation methods may produce biased inference. This is particularly relevant in interconnected economies like Indian states.

Hausman (1978) in “Specification Tests in Econometrics.” The Hausman test determines whether Fixed Effects or Random Effects is appropriate. The null hypothesis favors Random Effects, assuming no correlation between unobserved effects and explanatory variables. Rejection implies that Fixed Effects is more appropriate, as it accounts for such correlations. This ensures consistent and unbiased estimation.

Toda and Yamamoto (1995), who proposed a modified causality testing approach that avoids pre-testing biases. Their method allows valid causality inference even when variables are integrated, making it robust for macroeconomic data.

Damodar Gujarati and Jeffrey Wooldridge, provide the theoretical basis for regression analysis. These works explain model assumptions, estimation techniques, and the consequences of violations such as heteroskedasticity and autocorrelation.

Driscoll and Kraay (1998), in “Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data,” address the problem of unreliable standard errors in panel models. Traditional methods assume no cross-sectional dependence, homoskedasticity, and no autocorrelation assumptions often violated in practice. The Driscoll-Kraay method provides robust standard errors that remain valid under cross-sectional dependence, heteroskedasticity, and serial correlation, ensuring reliable statistical inference.

2.3 Review of Studies with Reference to Indian States

Mishra, Rout and Mohapatra (2011), “Causality between Tourism and Economic Growth: Empirical Evidence from India,” examines the causal relationship between tourism and economic growth at the national level. Using GDP and tourism indicators such as tourist arrivals and receipts, the study applies Johansen cointegration and Vector Error Correction Model (VECM). The findings reveal unidirectional causality from tourism to economic growth, supporting the Tourism-Led Growth Hypothesis (TLGH). However, the study is limited by its aggregate approach, as it does not account for regional disparities across Indian states.

Dash, Parida and Sahoo (2014), in “Tourism and Economic Growth in India: Evidence from State-Level Panel Data,” uses state-level panel data with GSDP, Domestic Tourist Visits (DTV), and Foreign Tourist Arrivals (FTA). Applying Pooled OLS and Fixed Effects models, the study finds that tourism positively affects economic growth, but the magnitude varies significantly across states. The results highlight that infrastructure and policy differences determine the effectiveness of tourism, though the study lacks advanced econometric corrections.

Kumar and Kumar (2013), “Tourism and Economic Growth in India: A Panel Data Approach,” examines the long-run relationship between tourism and economic growth using panel cointegration and Fixed Effects models. The study finds a positive long-run relationship and shows that tourism operates alongside factors such as population and investment. However, it does not address econometric issues like cross-sectional dependence and heteroskedasticity, limiting robustness.

Suresh and Senthilnathan (2014), “Relationship between Tourism and Economic Growth in India: A Causality Analysis,” applies Granger causality techniques and finds a bidirectional relationship between tourism and economic growth. This indicates that tourism and economic development reinforce each other. However, the analysis is conducted at the national level, overlooking state-level differences.

Sharma and Dey (2015), “Tourism Development, Infrastructure and Economic Growth in India,” incorporates infrastructure variables along with tourism indicators. The findings show that tourism contributes to growth, but its effectiveness depends on infrastructure and public investment. This highlights that tourism alone is insufficient without supporting economic conditions.

Bhatia (2011), “Tourism Development: Principles and Practices,” provides a conceptual understanding of tourism’s role in employment generation, income creation, and regional development. The study emphasizes that tourism outcomes depend on state-level planning and infrastructure, reinforcing the importance of regional analysis.

Rao and Singh (2018), “Tourism and Regional Development in India,” examines regional disparities in tourism benefits. Using state-level data, the study finds that tourism growth is concentrated in a few developed states, indicating that tourism can reinforce regional inequalities rather than reduce them.

Kaur and Sharma (2017), “Tourism and Employment Generation in India,” shows that tourism significantly contributes to job creation in service sectors. This highlights an indirect channel through which tourism influences economic growth.

Narayana (2010), “Tourism and Economic Development in India,” finds that tourism positively impacts economic development, but the strength of this relationship depends on infrastructure and local economic conditions.

Saha and Yap (2014), “The Moderation Effects of Political Instability on Tourism Development and Economic Growth,” shows that tourism contributes to growth, but the effect is influenced by governance and political stability. This suggests that institutional quality plays a key role in determining tourism outcomes.

Singh and Kaur (2016), “Tourism Demand and Economic Growth in India,” finds that tourism demand positively affects economic growth. However, it also highlights that tourism is highly sensitive to external shocks, making it a volatile driver of growth.

2.4 Summary of the Literature Review

The review of literature on tourism and economic growth, at both global and Indian levels, indicates that tourism generally contributes positively to economic growth. However, the strength and direction of this relationship vary depending on economic conditions, regional characteristics, and methodological approaches. At the global level, studies such as Balaguer and Cantavella-Jordá (2002) provide strong evidence that tourism can drive long-run economic growth. Later research, including Lee and Chang (2008) and Narayan, Narayan and Prasad

(2010), shows that the tourism growth relationship differs across countries, with unidirectional or bidirectional effects depending on the level of development. The literature also emphasizes the importance of econometric rigor, with contributions from Im, Pesaran and Shin (2003), Kao (1999), Pesaran (2004), and Driscoll and Kraay (1998) highlighting the need to address issues such as non-stationarity, cross-sectional dependence, and heteroskedasticity. In the Indian context, early studies like Mishra, Rout and Mohapatra (2011) confirm that tourism contributes to economic growth at the national level. However, such approaches overlook regional differences. Later studies, including Dash, Parida and Sahoo (2014) and Kumar and Kumar (2013), adopt state-level panel data and show that tourism positively affects growth, but the impact varies across states due to differences in infrastructure and development. Further evidence from Suresh and Senthilnathan (2014) suggests a bidirectional relationship between tourism and economic growth, while studies such as Sharma and Dey (2015) and Bhatia (2011) emphasize the role of infrastructure and policy support. Research by Rao and Singh (2018) highlights regional inequalities, whereas Kaur and Sharma (2017) and Narayana (2010) show that tourism contributes indirectly through employment and regional development. Additionally, Saha and Yap (2014) demonstrate the influence of institutional factors. Despite these contributions, a major gap remains. Most Indian studies rely on basic econometric methods such as Pooled OLS and standard Fixed Effects models, without addressing issues like cross-sectional dependence, heteroskedasticity, and autocorrelation. Ignoring these factors can lead to biased and unreliable results. The literature leads to three key conclusions. First, tourism contributes to economic growth, particularly in developing economies. Second, its impact is not uniform and depends on regional conditions. Third, methodological rigor is essential to obtain reliable and meaningful results.

Table 2.1: Summary of Literature Review

Author(s) & Year	Objective, Country & Time	Methodology	Variables Used	Findings
Balaguer & Cantavella Jordá (2002)	Examine tourism led growth in Spain (long run)	Cointegration & Granger causality	GDP, Tourism receipts	Tourism positively drives economic growth in the long run

Lee & Chang (2008)	Analyze tourism growth nexus in OECD & non OECD countries	Panel cointegration & causality	GDP, Tourist arrivals	Tourism drives growth in OECD; bidirectional in developing countries
Narayan et al. (2010)	Study tourism impact in developing economies	Panel data analysis	GDP, Tourism earnings	Strong positive effect of tourism on growth
Cortes Jimenez & Pulina (2010)	Examine inbound tourism and growth	Cointegration analysis	GDP, Foreign tourism	Foreign tourism contributes positively but unevenly
Tang & Tan (2015)	Assess tourism effectiveness in Malaysia	Time series econometrics	GDP, Tourism variables	Tourism supports growth with proper infrastructure
Brida et al. (2016)	Review TLGH globally	Literature review	Multiple variables	Tourism growth relationship varies by region and method
Mishra, Rout & Mohapatra (2011)	Test tourism growth causality in India	Johansen cointegration & VECM	GDP, Tourist arrivals, Receipts	Tourism leads economic growth (unidirectional)
Dash, Parida & Sahoo (2014)	Analyze tourism growth at Indian state level	Panel OLS & Fixed Effects	GSDP, DTV, FTA	Positive but uneven impact across states
Kumar & Kumar (2013)	Examine long run tourism growth in India	Panel cointegration & FE	GSDP, Tourism, Population, Investment	Long run positive relationship exists

Suresh & Senthilnathan (2014)	Identify causality between tourism and growth in India	Granger causality	GDP, Tourist arrivals	Bidirectional relationship between tourism and growth
Sharma & Dey (2015)	Study role of infrastructure in tourism growth	Regression analysis	GDP, Tourism, Infrastructure	Infrastructure strengthens tourism's impact
Bhatia (2011)	Explain tourism development in India	Conceptual analysis	Tourism sector indicators	Tourism contributes to employment and regional development
Rao & Singh (2018)	Examine regional disparity in tourism growth in India	Regional analysis	Tourism & economic indicators	Benefits concentrated in few states
Kaur & Sharma (2017)	Analyze tourism's role in employment generation	Empirical analysis	Tourism, Employment	Tourism creates significant employment
Narayana (2010)	Study tourism and regional development in India	Economic analysis	Tourism flows, GDP	Tourism supports development but depends on local conditions
Saha & Yap (2014)	Analyze role of political stability in tourism growth	Panel regression	GDP, Tourism	Institutional quality affects tourism effectiveness
Singh & Kaur (2016)	Examine tourism demand and growth in India	Time series analysis	Tourist arrivals, GDP	Tourism demand drives growth but is volatile

2.5 Research Gap

The existing literature provides substantial evidence that tourism contributes to economic growth. However, several limitations remain in both global and Indian studies. First, many studies rely on outdated data and fail to capture recent structural changes in the Indian economy, including digitalization, infrastructure expansion, and disruptions such as the COVID-19 pandemic. This limits the relevance of earlier findings in the current context. Second, a significant portion of Indian literature, including studies such as Mishra, Rout and Mohapatra (2011) and Suresh and Senthilnathan (2014), adopts a national-level approach, treating India as a homogeneous economy. This overlooks substantial regional differences across states in terms of infrastructure, policy, and economic development. Third, although later studies such as Dash, Parida and Sahoo (2014) and Kumar and Kumar (2013) employ state-level panel data, they rely on basic econometric models without adequately addressing key issues such as cross-sectional dependence, heteroskedasticity, and autocorrelation. Ignoring these factors can lead to biased and unreliable estimates. Fourth, tourism is often analyzed in isolation, with limited consideration of supporting factors such as infrastructure and government investment, despite evidence from studies like Sharma and Dey (2015) that these factors significantly influence tourism outcomes. Given these limitations, this study addresses the gap by using recent panel data for 15 Indian states from 2011 to 2023, incorporating both domestic and foreign tourism along with control variables such as population and capital outlay. It adopts a robust econometric framework, including unit root and cointegration testing, and applies a Fixed Effects model with Driscoll-Kraay standard errors to correct for cross-sectional dependence, heteroskedasticity, and autocorrelation.

CHAPTER 3

DATA AND METHODOLOGY

3.1 Introduction

This chapter outlines the data, variables and econometric methods used to examine the relationship between tourism development and economic growth across Indian states. The study adopts a panel data approach combining information across states and over time which allows the analysis to capture both regional differences and changes over the years. Using panel data also helps control for state-specific factors that are not directly observable but may influence economic outcomes, thereby improving the reliability of the results. The analysis is based on a balanced panel dataset covering 15 states from 2011 to 2023 resulting in a total of 195 observations. This chapter provides the methodological framework of the study including the data sources definition of variables, and the econometric techniques used for estimation.

3.2 Source of Data

This study uses secondary data collected from official government sources to ensure reliability and consistency across states and over time. Gross State Domestic Product (GSDP) which represents economic growth is obtained from the Ministry of Statistics and Programme Implementation. Tourism-related variables, Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA) are sourced from the Ministry of Tourism. Population (POP) data is collected from the Census of India, while Capital Outlay (CAP), representing government investment, is obtained from the Reserve Bank of India. And a COVID-19 dummy variable is constructed to capture the impact of the pandemic taking the value 1 during the COVID period and 0 otherwise.

Table 3.1: Variables and Sources

Variable	Description	Source
GSDP	Gross Net Domestic product	Ministry of Statistics programme Implementation
DTV	Domestic Tourist Visits	Ministry of Tourism
FTA	Foreign Tourist Arrivals	Ministry of Tourism
POP	Population	Census of India
CAP	Capital Outlay	Reserve Bank of India
COVID	Dummy Variable	Constructed

3.3 Description of Variables

This study uses one dependent variable and several independent variables to examine the relationship between tourism and economic growth. The dependent variable is Gross State Domestic Product (GSDP), which measures economic growth. It is used in logarithmic form ($\ln_gsdp$) to improve comparability and interpretation. The independent variables include Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA), which represent tourism activity. Population (POP) is included as a control variable to capture labour supply and market size. Capital Outlay (CAP) is used to represent government investment in infrastructure. All continuous variables are transformed into their natural logarithmic form. This allows the coefficients to be interpreted as elasticities, meaning they show the percentage change in economic growth resulting from a one percent change in the explanatory variables.

3.4 Measurement of Variables

This section explains how the variables are defined and transformed for econometric analysis. All continuous variables are converted into their natural logarithmic form to ensure comparability across states and to improve interpretation. This transformation helps reduce scale differences, minimize heteroskedasticity, and allows the coefficients to be interpreted as elasticities.

The dependent variable of the study is Gross State Domestic Product (GSDP), which represents the level of economic activity in each state. It is transformed as:

$$\ln(GSDP)_{it} \quad 3.1$$

This transformation implies that the model explains the percentage change in economic growth rather than absolute changes.

The explanatory variables of tourism are Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA). These are expressed as:

$$\ln(DTV)_{it}, \ln(FTA)_{it} \quad 3.2$$

These variables capture the intensity of tourism activity. By taking logarithms, the model measures how percentage changes in tourism flows influence economic growth.

To control for internal economic strength, the study includes Population (POP), which is specified

$$\ln(POP)_{it} \quad 3.3$$

Population acts as a proxy for labour supply and market size. Its inclusion ensures that the model does not wrongly attribute economic growth solely to tourism.

Another important variable is Capital Outlay (CAP), representing government investment in infrastructure and development. It is defined as:

$$\ln (CAP)_{it} \quad 3.4$$

This variable captures the role of public expenditure in supporting economic activity and tourism development.

In addition to these continuous variables, the study incorporates a dummy variable to capture the structural impact of the COVID-19 pandemic. This is defined as:

$$COVID_{it} = \begin{cases} 1, & \text{for COVID periods} \\ 0, & \text{otherwise} \end{cases} \quad 3.5$$

Unlike the other variables, this dummy is not log-transformed. It is used to capture the shift in economic conditions during the pandemic period.

Since the model is specified in log-log form, the coefficients of the explanatory variables (DTV, FTA, POP, CAP) are interpreted as elasticities. A 1% increase in Domestic Tourist Visits (DTV) leads to a $\beta_1\%$ change in GSDP, A 1% increase in Foreign Tourist Arrivals (FTA) leads to a $\beta_2\%$ change in GSDP, A 1% increase in Population (POP) leads to a $\beta_3\%$ change in GSDP, A 1% increase in Capital Outlay (CAP) leads to a $\beta_4\%$ change in GSDP

For the COVID dummy variable, the interpretation is different. Since it is not in logarithmic form, its effect is interpreted as:

$$\% \Delta = (e^{\beta_5} - 1) \times 100 \quad 3.6$$

This represents the percentage change in GSDP during the COVID period relative to the non-COVID period.

3.5 Model Specification

This study examines the relationship between tourism development and economic growth across Indian states using a panel data framework. The model incorporates tourism indicators along with control variables to capture broader economic influences.

The functional relationship is expressed as:

$$GSDP_{it} = f(DTV_{it}, FTA_{it}, POP_{it}, CAP_{it}, COVID_{it}) \quad 3.7$$

To improve interpretability and reduce heteroskedasticity, all continuous variables are transformed into natural logarithmic form. The econometric model is specified as:

$$\ln GSDP_{it} = \beta_0 + \beta_1 \ln DTV_{it} + \beta_2 \ln FTA_{it} + \beta_3 \ln POP_{it} + \beta_4 \ln CAP_{it} + \beta_5 COVID_{it} + u_{it} \quad 3.8$$

Where:

- $i = 1, 2, \dots, N$ represents states
- $t = 1, 2, \dots, T$ represents time (2011- 2023)
- $\ln\text{GSDP}$ = Log of Gross State Domestic Product
- $\ln\text{DTV}$ = Log of Domestic Tourist Visits
- $\ln\text{FTA}$ = Log of Foreign Tourist Arrivals
- $\ln\text{POP}$ = Log of Population
- $\ln\text{CAP}$ = Log of Capital Outlay
- COVID = Dummy variable (1 = pandemic period, 0 otherwise)
- u_{it} = Error term

The coefficients $\beta_1, \beta_2, \beta_3$, and β_4 represent elasticities, indicating the percentage change in GSDP resulting from a 1% change in the respective explanatory variables. The coefficient β_5 captures the average percentage difference in GSDP during the COVID period relative to the non-COVID period.

Panel Model Extensions

To account for the panel structure of the data, the study estimates three alternative models: Pooled Ordinary Least Squares (OLS), Fixed Effects (FE), and Random Effects (RE).

Pooled OLS Model

The Pooled OLS model assumes that there are no differences across states and estimates a common relationship for all observations:

$$\ln\text{GSDP}_{it} = \beta_0 + \beta_1 \ln\text{DTV}_{it} + \beta_2 \ln\text{FTA}_{it} + \beta_3 \ln\text{POP}_{it} + \beta_4 \ln\text{CAP}_{it} + \beta_5 \text{COVID}_{it} + u_{it} \quad 3.9$$

This model treats the dataset as a single pooled sample and ignores unobserved state-specific heterogeneity.

Fixed Effects Model (FE)

The Fixed Effects model accounts for unobserved, time-invariant characteristics of each state by allowing a state-specific intercept:

$$\ln\text{GSDP}_{it} = \alpha_i + \beta_1 \ln\text{DTV}_{it} + \beta_2 \ln\text{FTA}_{it} + \beta_3 \ln\text{POP}_{it} + \beta_4 \ln\text{CAP}_{it} + \beta_5 \text{COVID}_{it} + u \quad 3.10$$

Where:

- α_i represents state-specific effects capturing unobserved characteristics such as geography, institutional quality, and infrastructure

This model focuses on within-state variation over time and provides consistent estimates when unobserved effects are correlated with explanatory variables.

Random Effects Model (RE)

The Random Effects model assumes that state-specific effects are random and uncorrelated with the explanatory variables:

$$\ln GSDP_{it} = \beta_0 + \beta_1 \ln DTV_{it} + \beta_2 \ln FTA_{it} + \beta_3 \ln POP_{it} + \beta_4 \ln CAP_{it} + \beta_5 COVID_{it} + u_i + \epsilon_{it} \quad 3.11$$

Where:

- μ_i = Random state-specific effect
- ϵ_{it} = Idiosyncratic error term

The Random Effects model utilizes both within-state and between-state variation, but its validity depends on the assumption that μ_i is not correlated with the explanatory variables.

Model Selection

To determine the appropriate model, the Hausman specification test is employed. The test evaluates whether the Random Effects assumption holds.

- **Null Hypothesis (H_0):** Random Effects model is appropriate (no correlation)
- **Alternative Hypothesis (H_1):** Fixed Effects model is appropriate (correlation exists)

If the null hypothesis is rejected, the Fixed Effects model is preferred as it provides consistent and reliable estimates.

3.5 Pre-Diagnostic Tests

Before estimating the model, it is necessary to check whether the data satisfies the required conditions for econometric analysis. In panel data, issues such as non-stationarity, multicollinearity and lack of long-run relationships can affect the validity of the results. Therefore, pre-diagnostic tests are conducted to examine the stability of variables, the presence of long-run relationships, and the correlation among explanatory variables. These tests ensure

that the data is suitable for regression analysis and help in selecting the appropriate econometric model.

3.5.1 Multicollinearity Test (Variance Inflation Factor - VIF)

One of the first concerns in regression analysis is whether the explanatory variables are excessively correlated with each other. When two or more independent variables move closely together, it becomes difficult to distinguish their individual impact on the dependent variable. This issue is referred to as multicollinearity, and if it is present at a high level, it can distort the estimated coefficients and make the results unstable. In this study, the presence of multicollinearity is examined using the Variance Inflation Factor (VIF). The VIF measures how much the variance of an estimated coefficient increases due to the correlation among explanatory variables. It is defined as:

$$VIF = \frac{1}{1-R^2} \quad 3.12$$

Where,

- VIF = Variance Inflation Factor for the independent variable
- R^2 = Coefficient of determination obtained by regressing the independent variable on all other independent variables

If $VIF=1 \rightarrow$ No multicollinearity, If $VIF < 5 \rightarrow$ Acceptable, If $VIF > 5 \rightarrow$ Potential problem, If $VIF > 10 \rightarrow$ Serious multicollinearity. A higher value of VIF indicates a greater degree of multicollinearity. As a general rule, a VIF value greater than 5 (or 10 in stricter cases) signals a potential problem. If multicollinearity is present, it can lead to inflated standard errors and unreliable coefficient estimates. By conducting the VIF test, the study ensures that each explanatory variable such as tourism indicators, population, and capital outlay contributes uniquely to explaining economic growth, thereby improving the reliability and interpretability of the regression results.

3.5.2 Panel Unit Root Test (Im-Pesaran-Shin Test)

Another critical issue in time-series and panel data analysis is the presence of non-stationarity. A variable is said to be non-stationary if its statistical properties such as mean and variance change over time. Using non-stationary data in regression can lead to spurious results where relationships appear statistically significant even though no meaningful economic relationship exists. To address this issue the study employs the Im-Pesaran-Shin (IPS) panel unit root test which is particularly suitable for panel datasets with heterogeneity across cross-sectional units

such as states. Unlike traditional unit root tests the IPS test allows each state to have its own dynamic behaviour.

The general form of the test is given by:

$$\Delta Y_{it} = \alpha_i + \beta_i Y_{it-1} + \epsilon_i \quad 3.13$$

Where:

- Y_{it} = Value of the variable (such as $\ln_gdp$, $\ln_dtv$, $\ln_fta$, $\ln_pop$, $\ln_cap$) for state i at time t
- i = Cross-sectional unit (states like Tamil Nadu, Kerala, etc.)
- t = Time period (years from 2011 to 2023)
- ΔY_{it} = Change in the variable (i.e., $Y_{it} - Y_{it-1}$)
- α_i = State-specific constant term (captures individual state effects)
- β_i = Coefficient measuring the influence of past values on current change
- Y_{it-1} = Lagged value of the variable (previous year value)
- ϵ_i = Error term (random shocks or unexplained factors)

The null hypothesis states that all panels contain a unit root (i.e., the data is non-stationary), while the alternative hypothesis suggests that at least some panels are stationary. The importance of this test lies in its ability to determine whether the variables used in the model are stable over time. If the variables are non-stationary further steps such as cointegration testing are required to ensure that the regression results are valid.

3.5.3 Panel Cointegration Test (Kao Test)

When the variables in the study are found to be non-stationary, the next step is to examine whether they share a long-run equilibrium relationship. This is important because even if individual variables are non-stationary, they may still move together over time in a stable manner. This concept is known as cointegration. To test for this, the study uses the Kao panel cointegration test, which is specifically designed for panel data analysis. The test assumes a common long-run relationship across all cross-sectional units.

The model can be expressed as:

$$Y_{it} = \alpha_i + \beta X_{it} + \epsilon_{it} \quad 3.14$$

Where:

- Y_{it} = Dependent variable ($\ln_gdp$) for state i at time t
- X_{it} = Vector of independent variables ($\ln_dtv$, $\ln_fta$, $\ln_pop$, $\ln_cap$, $covid$) for state i at time t
- i = Cross-sectional units (states)
- t = Time period (years from 2011 to 2023)
- α_i = State-specific constant (captures unobserved heterogeneity across states)

- β = Long-run coefficient measuring the relationship between independent variables and economic growth
- ϵ_{it} = Error term (residuals representing deviations from long-run equilibrium)

The null hypothesis of the Kao test is that there is no cointegration among the variables, while the alternative hypothesis suggests the existence of a long-run relationship. If cointegration is confirmed it implies that the variables such as tourism activity economic growth population and capital outlay move together over time despite short-run fluctuations. This ensures that the estimated regression model is not spurious and that the relationships identified are economically meaningful. Thus the cointegration test acts as a crucial bridge between non-stationary data and valid regression analysis allowing the study to proceed with confidence in the existence of a stable long-run relationship.

3.6 Main Model and Estimation Strategy

After confirming that the data satisfies the required conditions, the study proceeds to estimate the relationship between tourism development and economic growth. Since panel data combines information across states and time, a single model is not sufficient to capture these complexities. Therefore, the study follows a step-by-step estimation approach, starting with a basic model and then moving to more advanced models that account for differences across states. This approach allows for a better understanding of the data, helps control for unobserved state-specific effects, and ensures that the final model selected is both statistically reliable and economically meaningful.

3.6.1 Pooled Ordinary Least Squares (Pooled OLS)

The estimation process begins with the most basic approach Pooled Ordinary Least Squares. At this stage, all observations from different states and different time periods are combined into a single dataset, and a common relationship is estimated without distinguishing between individual states.

$$Y_{it} = \beta_0 + \beta X_{it} + \epsilon_{it} \quad 3.15$$

Where:

- Y_{it} =Dependent variable (ln_gsdp) for state i at time t
- X_{it} =Vector of independent variables (ln_dtv, ln_fta, ln_pop, ln_cap, covid)
- I =Cross-sectional units (states)
- T =Time period (2011 - 2023)
- β_0 =Intercept term (common for all states)
- β =Coefficient measuring the impact of explanatory variables on economic growth

- ϵ_{it} =Error term (captures unexplained variations)

At first glance, this approach appears convenient, as it treats the dataset as a unified whole and provides a quick understanding of the relationship between tourism and economic growth. However, this simplicity comes at a cost. By ignoring differences across states, the model implicitly assumes that all states are identical in terms of their economic structure, institutional framework, and tourism potential. In reality, this assumption is unrealistic. Indian states differ significantly in terms of infrastructure, policy environment, and cultural attractiveness. As a result, the Pooled OLS model fails to capture these underlying differences, leading to biased and unreliable estimates. Thus, while it serves as a useful starting point, it cannot be considered an appropriate model for panel data analysis.

3.6.2 Fixed Effects Model (FE)

Recognizing the limitations of the Pooled OLS approach, the analysis moves to a more refined model the Fixed Effects model. Unlike the previous model, the Fixed Effects approach explicitly acknowledges that each state has its own unique characteristics that do not change over time.

$$Y_{it} = \alpha_i + \beta X_{it} + \epsilon_{it} \quad 3.16$$

Where:

- Y_{it} =Dependent variable (ln_gsdg) for state i at time t
- X_{it} =Vector of independent variables (ln_dtv, ln_fta, ln_pop, ln_cap, covid)
- I=Cross-sectional units (states)
- T=Time period (2011-2023)
- α_i =State-specific intercept (captures unobserved characteristics unique to each state)
- β =Coefficient measuring the impact of explanatory variables on economic growth
- ϵ_{it} =Error term

Behind the Fixed Effects model is that each state possesses certain characteristics such as geography cultural heritage infrastructure quality and institutional framework that remain constant over time but significantly influence economic performance. These factors are not directly observable in the dataset yet ignoring them can lead to biased results. Unlike the Pooled OLS model which assumes all states are identical the Fixed Effects model allows each state to have its own intercept through. This effectively controls for all time invariant differences across states ensuring that the estimated relationship reflects only the variation within each state over time. An important assumption of this model is that these unobserved state-specific effects may

be correlated with the explanatory variables. For instance a state with better infrastructure may attract more tourists and also experience higher economic growth. If such correlation is ignored, the estimated coefficients would be biased. By accounting for these unobserved but important factors the Fixed Effects model provides more reliable and realistic estimates. It isolates the true impact of tourism and other variables on economic growth making it a more appropriate choice for panel data analysis.

3.6.3 Random Effects Model (RE)

As an alternative to the Fixed Effects approach, the Random Effects model is also considered. While it accounts for differences across states it treats these differences as random variations rather than fixed parameters

$$Y_{it} = \alpha + \beta X_{it} + u_i + \epsilon_{it} \quad 3.17$$

Where:

- Y_{it} =Dependent variable (ln_gsdp) for state i at time t
- X_{it} =Vector of independent variables (ln_dtv, ln_fta, ln_pop, ln_cap, covid)
- I=Cross-sectional units (states)
- T=Time period (2011 - 2023)
- α =Common intercept
- β =Coefficient measuring the impact of explanatory variables
- u_i =Random state-specific effect (captures unobserved heterogeneity across states)
- ϵ_{it} =Error term

The Random Effects model also recognizes that differences exist across states but instead of treating these differences as fixed and unique it assumes that they are randomly distributed across the panel. In this framework the state-specific component captures unobserved factors such as regional characteristics institutional differences and infrastructure variations.

The assumption of the Random Effects model is that these unobserved state-specific effects are not correlated with the explanatory variables. This assumption allows the model to utilize both within-state variation (over time) and between-state variation (across states) making it more efficient than the Fixed Effects model if the assumption holds true.

However, this assumption is quite strong and often unrealistic in practical situations. In the context of this study, it is likely that unobserved factors such as infrastructure quality or governance are related to tourism activity and economic growth. If such correlation exists, the Random Effects model produces inconsistent and biased estimates.

Therefore while the Random Effects model is considered as part of the estimation strategy, its validity must be tested before it can be accepted.

3.6.4 Hausman Test (Model Selection)

After estimating both the Fixed Effects and Random Effects models it becomes necessary to determine which model is more appropriate for the data. This decision is made using the Hausman specification test which evaluates whether the assumptions of the Random Effects model are valid.

$$\mathbf{H} = (\mathbf{b} - \mathbf{B})' (\mathbf{V}_b - \mathbf{V}_B) (\mathbf{b} - \mathbf{B}) \quad 3.18$$

Where:

- $\mathbf{H}$ = Hausman test statistic
- $\mathbf{b}$ = Coefficient estimates from the Fixed Effects model
- $\mathbf{B}$ = Coefficient estimates from the Random Effects model
- $\mathbf{V}_b$ = Variance - covariance matrix of Fixed Effects estimates
- $\mathbf{V}_B$ = Variance - covariance matrix of Random Effects estimates
- $(\mathbf{b} - \mathbf{B})$ = Difference between FE and RE estimates

The Hausman test is designed to resolve the key question: Should the Fixed Effects model or the Random Effects model be used? The answer depends on whether the fundamental assumption of the Random Effects model holds true. The Random Effects model assumes that the unobserved state-specific effects are not correlated with the explanatory variables. If this assumption is valid both Fixed Effects and Random Effects models should produce similar coefficient estimates. However if there is a significant difference between the two sets of estimates, it indicates that the Random Effects assumption is violated. In such a case the Fixed Effects model becomes more appropriate because it accounts for this correlation and provides consistent estimates. The Hausman test formally evaluates this difference and provides a statistical basis for model selection.

Decision Rule

Null Hypothesis (H_0): Random Effects model is appropriate (no correlation)

Alternative Hypothesis (H_1): Fixed Effects model is appropriate (correlation exists)

- If $p\text{-value} < 0.05 \rightarrow$ Reject $H_0 \rightarrow$ Use Fixed Effects
- If $p\text{-value} > 0.05 \rightarrow$ Use Random Effects

The Hausman test result is statistically significant, leading to the rejection of the null hypothesis. This indicates that the Random Effects model is not appropriate, and the Fixed Effects model is selected for the analysis. The validity of the selected model depends on several assumptions, including the absence of multicollinearity, stationarity of variables, existence of long-run relationships, homoskedasticity, no autocorrelation, and cross-sectional independence. These assumptions are tested using pre-diagnostic and post-diagnostic tests. The pre-diagnostic tests ensure that the data is suitable for estimation, while the post-diagnostic tests check the reliability of the model. Based on these results, appropriate corrections are applied to obtain robust and valid estimates.

3.7 Post-Diagnostic Tests and Model Correction

After selecting the Fixed Effects model, it is necessary to check whether the results are reliable. In panel data, issues such as heteroskedasticity, autocorrelation, and cross-sectional dependence can affect the accuracy of standard errors and lead to misleading conclusions. Therefore, post-diagnostic tests are conducted to identify these problems. If such issues are present, appropriate corrections are applied to ensure that the estimated results are statistically reliable and valid.

3.7.1 Heteroskedasticity Test (Modified Wald Test)

One of the key assumptions of classical regression models is that the variance of the error term remains constant across observations. When this assumption is violated the problem is known as heteroskedasticity. In panel data this issue often arises because different states have different economic sizes levels of development and variability in growth. To detect this problem, the study employs the Modified Wald test for groupwise heteroskedasticity in the Fixed Effects model. H_0

$$H_0 : \sigma_i^2 = \sigma^2 \quad \forall_i \quad 3.19$$

Where:

H_0 = Null hypothesis (homoskedasticity)

i = Variance of the error term for state i

σ_i^2 = Constant variance across all states

σ^2 = Cross-sectional units (states)

$\forall_i$ = For all states in the panel

The null hypothesis assumes that the variance of the error term is constant across all states, meaning that the model satisfies the assumption of homoskedasticity. The alternative

hypothesis suggests that the variance differs across states indicating the presence of heteroskedasticity. In practical terms heteroskedasticity implies that some states exhibit more variability in economic growth than others. For example economically larger or more volatile states may show greater fluctuations compared to smaller or more stable states. If heteroskedasticity is present the estimated standard errors become unreliable, which in turn affects the accuracy of t-statistics and p-values. As a result conclusions about the statistical significance of variables may be incorrect.

3.7.2 Autocorrelation Test (Wooldridge Test)

Another important assumption of regression analysis is that the error terms are not correlated over time. When the error terms of a variable are related to their past values the problem is known as autocorrelation. In panel data this issue commonly arises because economic conditions in one period often influence outcomes in subsequent periods. To detect this problem the study employs the Wooldridge test for autocorrelation in panel data which is specifically designed for fixed effects models.

$$Y_{it} = \rho Y_{it-1} + \epsilon_{it} \quad 3.20$$

The Wooldridge test examines whether the error terms in the model are correlated across time within each state. The null hypothesis assumes that there is no autocorrelation meaning that the error term in one period is independent of the error term in the previous period. However, in real-world economic data this assumption is often violated. For example if a state experiences economic growth or decline in one year it is likely that similar patterns will continue in the following year. This persistence creates correlation in the error terms over time. If autocorrelation is present it does not necessarily bias the coefficient estimates but it leads to incorrect standard errors. This results in unreliable t-statistics and p-values which can mislead the interpretation of statistical significance.

3.7.3 Cross-Section Dependence Test (Pesaran CD Test)

In panel data analysis an important assumption is that cross-sectional units (states) are independent of each other. However in reality this assumption is often violated especially in a country like India where states are economically and socially interconnected. This issue is known as cross-section dependence where the error terms of one state are influenced by those of another. To detect this problem the study employs the Pesaran Cross-Section Dependence (CD) test which is widely used in panel data analysis.

$$CD = \frac{\sqrt{2T}}{N(N-1)} \sum \rho_{ij} \quad 3.21$$

The Pesaran CD test examines whether the residuals from different states are correlated with each other. The null hypothesis assumes that there is no cross-section dependence meaning that each state operates independently in terms of the error structure.

However this assumption is often unrealistic in practical scenarios. For instance, tourism development in one state may influence neighboring states through spillover effects shared infrastructure or common economic policies. Similarly macroeconomic shocks and national level events can simultaneously affect multiple states.

If cross section dependence is present it violates the assumption of independence across units. This leads to biased standard errors and invalid statistical inference making the results unreliable.

3.7.4 Model Correction: Driscoll-Kraay Standard Errors

The results of the post-diagnostic tests clearly indicate the presence of heteroskedasticity autocorrelation and cross-section dependence in the model. These issues violate the fundamental assumptions of classical regression analysis and if left unaddressed can lead to unreliable statistical inference. In particular the standard errors obtained from a conventional Fixed Effects model become biased which directly affects the validity of t-statistics and p values. To overcome these limitations the study employs Driscoll-Kraay standard errors within the Fixed Effects framework. Unlike conventional estimation techniques this method is specifically designed for panel data settings where cross-sectional units are interdependent and where error terms may exhibit both time-series and cross-sectional correlation.

$$\text{Var}^{\wedge}(\beta^{\wedge})_{DK} = (X'X)^{-1} \Omega^{\wedge}_{DK} (X'X)^{-1} \quad 3.22$$

Where:

$\text{Var}^{\wedge}(\beta^{\wedge})_{DK} \rightarrow$ Robust variance-covariance matrix

$X \rightarrow$ Matrix of independent variables

$\Omega^{\wedge}_{DK} \rightarrow$ Driscoll-Kraay estimator

The Driscoll-Kraay method modifies the standard errors of the estimated coefficients to make them robust against multiple forms of model violations. Specifically, it adjusts for:

- Heteroskedasticity $\rightarrow$ Unequal variance of errors across states
- Autocorrelation $\rightarrow$ Correlation of errors over time within the same state
- Cross-section dependence $\rightarrow$ Correlation of errors across different states

This makes it particularly suitable for panel data involving interconnected units such as Indian states, where economic and tourism activities are not independent but influenced by common

shocks and spillover effects. Unlike traditional robust standard errors which typically address only one issue (such as heteroskedasticity) the Driscoll-Kraay approach provides a comprehensive correction. It ensures that the estimated standard errors remain valid even when multiple violations occur simultaneously. Importantly this method does not alter the estimated coefficients of the model. Instead it improves the reliability of the inference by correcting the standard errors. As a result the significance levels of the variables are more accurately estimated leading to more credible conclusions.

CHAPTER 4

EMPIRICAL RESULTS AND DISCUSSION

4.1.1 Model Framework for Estimation

The empirical analysis presented in this chapter is based on the econometric model specified in Section 3.5. The model examines the impact of tourism development and related economic factors on Gross State Domestic Product (GSDP) across Indian states using panel data.

The estimated log-linear model is expressed as:

$$\ln GSDP_{it} = \beta_0 + \beta_1 \ln DTV_{it} + \beta_2 \ln FTA_{it} + \beta_3 \ln POP_{it} + \beta_4 \ln CAP_{it} + \beta_5 COVID_{it} + u_{it} \quad 4.1$$

Where i denotes the state and t denotes the time period. The model includes tourism indicators Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA) along with control variables such as Population (POP) and Capital Outlay (CAP). A dummy variable is included to capture the structural impact of the COVID-19 pandemic. The coefficients of the log-transformed variables represent elasticities, indicating the percentage change in GSDP associated with a one percent change in the respective explanatory variables, holding other factors constant. To estimate this relationship, three panel data models are employed: Pooled Ordinary Least Squares (OLS), Fixed Effects (FE), and Random Effects (RE). The Hausman specification test is used to determine the most appropriate model. Based on the test results, the Fixed Effects model is selected as it accounts for unobserved state-specific heterogeneity. Further, diagnostic tests reveal the presence of heteroskedasticity, autocorrelation, and cross-sectional dependence. To address these issues, Driscoll-Kraay standard errors are applied to obtain robust and reliable estimates. The results of these models are presented and discussed in the following sections

4.1 Introduction

This chapter presents the empirical results and analysis of the study, focusing on the relationship between tourism development and economic growth across Indian states. The analysis is based on panel data for 15 states over the period 2011 to 2023, capturing both cross-sectional and time-based variations. The chapter follows the methodological framework outlined in Chapter 3 to maintain consistency. It begins with descriptive statistics and correlation analysis to provide an initial understanding of the data. This is followed by the results of pre-diagnostic tests, including multicollinearity, stationarity, and cointegration, to ensure that the data is suitable for econometric analysis. Next, the results of panel data models

Pooled OLS, Fixed Effects, and Random Effects are presented. The Hausman test is used to select the appropriate model. Post-diagnostic tests are then conducted to check for heteroskedasticity, autocorrelation, and cross-sectional dependence. Since these issues are present, Driscoll-Kraay standard errors are applied to obtain robust estimates. Finally, the chapter interprets the corrected results and discusses the findings in the context of tourism-led economic growth.

4.2 Descriptive Statistics

Descriptive statistics provide an overview of the distribution and key characteristics of the variables used in the analysis. They help in understanding the central tendency, variability, and range of the data, which is essential before proceeding to further econometric estimation. Table 4.1 presents the summary statistics for all variables included in the study.

Table 4.1: Descriptive Statistics of Variables

Variable	Obs	Mean	Std. Dev.	Min	Max
ln_gsdp	195	17.83815	0.6447664	16.52966	19.31073
ln_fta	195	12.58433	2.225176	2.079442	15.74214
ln_dtv	195	17.41081	1.296309	12.21874	20.09937
ln_pop	195	17.92877	0.6040068	17.04833	19.27802
ln_cap	195	14.06072	0.9000893	11.18527	15.96327
Covid	195	0.1538462	0.3617299	0	1

(Source: Author's calculation using panel dataset of Indian states, 2011- 2023)

The dataset consists of 195 observations confirming a balanced panel structure across 15 Indian states over the study period. This balanced nature enhances the reliability of panel data estimation by ensuring consistency across both cross-sectional and time dimensions.

The mean value of the log of Gross State Domestic Product is 17.84 with a relatively low standard deviation of 0.64. This indicates that economic output across states is fairly stable although the observed range suggests moderate variation in economic performance.

The log of Foreign Tourist Arrivals exhibits a high standard deviation of 2.23 indicating substantial variation across states. This suggests that foreign tourism is highly concentrated in certain states reflecting unequal distribution of international tourist inflows.

The log of Domestic Tourist Visits shows a standard deviation of 1.29 indicating noticeable variation across states though less pronounced than foreign tourist arrivals. This implies that domestic tourism is more evenly distributed but still varies significantly across regions.

The log of Population has a relatively low standard deviation of 0.60, indicating stability in population distribution across states in logarithmic terms. This is consistent with the gradual nature of demographic changes.

The log of Capital Outlay shows moderate variability, with a standard deviation of 0.90, reflecting differences in infrastructure investment and development levels across states.

The COVID-19 dummy variable (covid) has a mean value of 0.1538, indicating that approximately 15% of the observations correspond to the pandemic period. This variable captures the structural disruption caused by COVID-19 on economic and tourism activities.

Note:

- **ln_gsdp**: Natural logarithm of Gross State Domestic Product
- **ln_fta**: Natural logarithm of Foreign Tourist Arrivals
- **ln_dtv**: Natural logarithm of Domestic Tourist Visits
- **ln_pop**: Natural logarithm of Population
- **ln_cap**: Natural logarithm of Capital Outlay
- **covid**: Dummy variable (1 = COVID-19 period, 0 = non-COVID period)

4.3 Correlation Analysis

Correlation analysis is used to examine the strength and direction of the linear relationship between variables. It also helps in identifying potential multicollinearity issues among the independent variables before proceeding to regression analysis.

Table 4.2: Correlation Matrix

Variable	ln_gsdp	ln_fta	ln_dtv	ln_pop	ln_cap	covid
ln_gsdp	1.0000					
ln_fta	0.5302	1.0000				
ln_dty	0.6120	0.6582	1.0000			
ln_pop	0.5286	0.3886	0.4159	1.0000		
ln_cap	0.4752	0.2505	0.3850	0.3426	1.0000	
Covid	0.1002	-0.3886	-0.3162	0.0272	-0.0262	1.0000

The correlation results indicate that all major explanatory variables are positively associated with economic growth though the strength of association varies across variables.

The log of Domestic Tourist Visits shows the strongest positive correlation with economic growth (0.6120) suggesting that domestic tourism plays a significant role in influencing state level economic performance. Similarly the log of Foreign Tourist Arrivals is moderately correlated with economic growth (0.5332) indicating the contribution of international tourism.

The log of Population also exhibits a moderate positive correlation (0.5286) reflecting the role of population size in supporting economic activity. The log of Capital Outlay shows a positive but relatively weaker correlation (0.4752) indicating that infrastructure investment contributes to growth though its effect may be indirect.

Among the independent variables the highest correlation is observed between $\ln_dtv$ and $\ln_fta$ (0.6582). While this indicates a relationship between domestic and foreign tourism it remains below the critical threshold of 0.80 suggesting that serious multicollinearity is not present.

The COVID dummy variable shows a weak positive correlation with economic growth (0.1002) and negative correlations with tourism variables ($\ln_fta$ and $\ln_dtv$). This reflects the adverse impact of the pandemic on tourism activity while its direct relationship with economic growth remains limited. since none of the pairwise correlations exceed the threshold level the model does not suffer from severe multicollinearity and the variables can be safely included in the regression analysis.

Note:

- **$\ln_gsdp$** : Natural logarithm of Gross State Domestic Product
- **$\ln_fta$** : Natural logarithm of Foreign Tourist Arrivals
- **$\ln_dtv$** : Natural logarithm of Domestic Tourist Visits
- **$\ln_pop$** : Natural logarithm of Population
- **$\ln_cap$** : Natural logarithm of Capital Outlay
- **covid**: Dummy variable (1 = COVID-19 period, 0 = non-COVID period)

4.4 Pre-Diagnostic Tests

Before proceeding with model estimation, it is essential to examine whether the dataset satisfies the required statistical conditions for panel data analysis. This section presents the results of pre diagnostic tests including multicollinearity stationarity and cointegration tests.

Table 4.3: Summary of Pre-Diagnostic Test Results

Diagnostic Test	Variable	Test Statistic	Decision Rule	P-value	Decision
VIF (Multicollinearity)	Mean VIF = 1.58	-	$VIF < 5$ acceptable	-	Accepted
IPS Unit Root Test	$\ln_gsdp$	3.8714	$p < 0.05 \rightarrow$ Stationary	0.9999	Non-Stationary
	$\ln_fta$	0.8129		0.2081	Non-Stationary

	ln_dtv	1.8761		0.0303	Stationary
	ln_pop	9.5950		0.0000	Stationary
	ln_cap	1.6205		0.9474	Non-Stationary
	Covid	2.6047		0.0046	Stationary
Kao Cointegration Test	ADF	3.8430	$p < 0.05 \rightarrow$ Cointegration	0.0001	Cointegration exists

(Source: Author's computation using Stata output)

The Variance Inflation Factor (VIF) test indicates a mean VIF value of 1.58, which is well below the threshold value of 5. This confirms the absence of multicollinearity among the explanatory variables.

The Im-Pesaran-Shin (IPS) unit root test results reveal a mixed order of integration among the variables. Specifically, ln_dtv, ln_pop, and Covid are found to be stationary at level, as their p-values are less than 0.05, leading to rejection of the null hypothesis of unit root. In contrast, ln_gsdp, ln_fta, and ln_cap are non-stationary at level, as their p-values exceed 0.05, indicating failure to reject the null hypothesis.

Given the presence of non-stationary variables, the Kao cointegration test was conducted to examine the existence of a long-run equilibrium relationship. The test result (ADF = 3.8430, $p = 0.0001$) leads to rejection of the null hypothesis of no cointegration. This confirms that the variables are cointegrated and share a stable long-run relationship. Therefore, despite the presence of non-stationary variables, the existence of cointegration justifies proceeding with panel regression models for further analysis.

4.5 Model Estimation and Selection

This section presents the results of different panel data models used to examine the relationship between tourism and economic growth. The study employs three estimation techniques:

- Pooled Ordinary Least Squares - treats all observations as a single dataset without considering differences across states.
- Fixed Effects Model - controls for unobserved, time-invariant characteristics specific to each state.
- Random Effects Model - assumes that state-specific effects are random and not correlated with the explanatory variables.

To identify the most appropriate model the Hausman test is used to compare the Fixed Effects and Random Effects models.

4.5.1 Pooled Ordinary Least Squares (OLS) Results

The estimation process begins with the Pooled Ordinary Least Squares (OLS) model, which treats all observations as a single combined dataset without accounting for unobserved state-specific heterogeneity. As such, this model serves as a baseline specification for comparison with more advanced panel models.

Table 4.4: Pooled OLS Estimation Results

Variable	Coefficient	Std. Error	t-Statistic	P-value
ln_dtv	0.1831	0.0323	5.66	0.0000
ln_fta	0.0853	0.0186	4.59	0.0000
ln_pop	0.1935	0.0569	3.40	0.0001
ln_cap	0.1477	0.0362	4.07	0.0000
Covid	0.5909	0.0911	6.48	0.0000

| $R^2 = 0.6086$ | $\text{Adj } R^2 = 0.5982$ | $F\text{-stat} = 58.77$ | $\text{Prob} > F = 0.0000$ |

(Source: Author's computation using Stata output)

The results indicate that all explanatory variables are statistically significant at the 1 percent level, as evidenced by their p-values being less than 0.01.

Since the model is specified in logarithmic form except for the Covid dummy, the estimated coefficients can be interpreted as elasticities. The coefficient of ln_dtv (0.1831) implies that a 1 percent increase in domestic tourist visits leads to an approximate 0.18 percent increase in GSDP, holding other factors constant. Similarly, ln_fta (0.0853) suggests that a 1 percent rise in foreign tourist arrivals increases GSDP by about 0.08 percent.

The coefficient of ln_pop (0.1935) indicates that population growth contributes positively to economic output, while ln_cap (0.1477) reflects the positive role of capital formation in driving economic growth.

The Covid dummy variable shows a positive and statistically significant coefficient (0.5909), suggesting a structural shift during the pandemic period. However, this result may appear counterintuitive, as the Covid-19 period was generally associated with economic contraction. This unexpected sign may be due to model limitations, omitted variables, or measurement issues, and should be interpreted with caution.

The model fit is reasonably strong, with an R^2 value of 0.6086, indicating that approximately 60.86 percent of the variation in GSDP is explained by the included variables. The F-statistic (58.77) and its associated probability value ($p < 0.01$) confirm that the model is jointly significant. Despite these results, the Pooled OLS model does not account for unobserved heterogeneity across states, which may lead to biased and inconsistent estimates. Therefore, more robust panel data models such as Fixed Effects and Random Effects are estimated in the subsequent sections.

4.5.2 Fixed Effects Model (FE) Results

The Fixed Effects (FE) model is employed to control for unobserved, time-invariant state-specific characteristics such as institutional quality, geography, and infrastructure. By focusing on within-state variation over time, the FE model provides more consistent and reliable estimates compared to the Pooled OLS model.

Table 4.5: Fixed Effects Estimation Results

Variable	Coefficient	Std . Error	t- Statistic	P- Value
ln_dtv	0.0594	0.0168	3.52	0.0001
ln_fta	-0.0277	0.0067	-4.15	0.0000
ln_pop	4.333	0.1808	23.96	0.0000
ln_cap	0.0184	0.0092	2.01	0.0046
Covid	-0.0156	0.0266	-0.59	0.0558
Constant	-60.8032	3.0811	-19.73	0.0000

| R^2 (within) = 0.8830 | F-stat = 264.16 | Prob > F = 0.0000 |

(Source: Author's computation using Stata output)

The results indicate that most variables remain statistically significant, although their magnitudes and signs differ notably from the Pooled OLS estimates. This highlights the importance of controlling for unobserved heterogeneity across states.

The coefficient of $\ln_dtv$ (0.0594) remains positive and significant, suggesting that a 1 percent increase in domestic tourist visits leads to a 0.06 percent increase in GSDP within a state, holding other factors constant. However, the magnitude is substantially lower than in the Pooled OLS model, indicating that part of the earlier effect was driven by cross-state differences rather than within-state variation.

Interestingly, $\ln_fta$ exhibits a negative and statistically significant coefficient (-0.0277), implying that increases in foreign tourist arrivals within a state are associated with a slight

decline in GSDP. This counterintuitive result may reflect structural factors such as tourism leakage, seasonal fluctuations, or uneven distribution of tourism benefits.

The coefficient of $\ln_pop$ (4.333) is positive and highly significant, indicating a strong association between population growth and economic output. However, the unusually large magnitude suggests potential model specification issues or the influence of underlying time trends, and therefore should be interpreted with caution.

The coefficient of $\ln_cap$ (0.0184) is positive and statistically significant, confirming the role of capital formation in promoting economic growth, although the effect size is relatively modest.

The Covid dummy variable is statistically insignificant at conventional levels ($p > 0.05$), indicating that after controlling for state-specific effects, the pandemic does not have a significant within-state impact on GSDP.

The model demonstrates a strong explanatory power, with a within R^2 value of 0.8830, indicating that approximately 88.30 percent of the variation in GSDP within states over time is explained by the included variables. The F-statistic (264.16) further confirms the significance of the model. The differences between Pooled OLS and Fixed Effects results suggest that ignoring state-specific heterogeneity can lead to biased estimates. The Fixed Effects model provides a more reliable framework for analyzing the relationship between tourism and economic growth across Indian states.

4.5.3 Random Effects Model (RE) Results

The Random Effects (RE) model is estimated as an alternative to the Fixed Effects model. Unlike the FE model, the RE model assumes that unobserved state-specific effects are random and uncorrelated with the explanatory variables. This assumption allows the inclusion of both within-state and between-state variation in the estimation.

Table 4.6: Random Effects Estimation Results

Variable	Coefficient	Std . Error	Z – Statistic	P- Value
$\ln_dtv$	0.1624	0.0238	6.81	0.0000
$\ln_fta$	-0.0542	0.0098	-5.48	0.0000
$\ln_pop$	2.2571	0.1915	11.78	0.0000
$\ln_cap$	0.0444	0.0139	3.18	0.0001
Covid	0.1336	0.0384	3.48	0.0001

Constant	-25.4207	3.2629	-7.79	0.0000
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| Wald $\chi^2 = 458.59$ | Prob > $\chi^2 = 0.0000$ |

(Source: Author's computation using Stata output)

The estimation results indicate that all variables are statistically significant at the 1 percent level. Similar to the previous models, the log-log specification implies that most coefficients can be interpreted as elasticities.

The coefficient of $\ln_dtv$ (0.1624) is positive and significant, suggesting that a 1 percent increase in domestic tourist visits leads to a 0.16 percent increase in GSDP. The magnitude is notably higher than in the Fixed Effects model, indicating that between-state variation contributes substantially to this relationship.

The coefficient of $\ln_fta$ (-0.0542) remains negative and significant, reinforcing the finding that increases in foreign tourist arrivals within states may not translate into proportional economic gains. This may reflect structural inefficiencies, tourism leakage, or uneven distribution of tourism revenue.

The coefficient of $\ln_pop$ (2.2571) is positive and highly significant, indicating a strong association between population and economic output. However, the magnitude remains relatively large, suggesting that population may be capturing broader structural or time-related effects, and thus should be interpreted with caution.

The coefficient of $\ln_cap$ (0.0444) is positive and significant, confirming the role of capital formation in promoting economic growth.

The Covid variable is positive and statistically significant (0.1336), implying that the pandemic period is associated with a positive shift in GSDP. However, this result contrasts with the Fixed Effects model, where the variable was insignificant, suggesting that the estimated impact of Covid may depend on model assumptions.

The model is statistically significant, as indicated by the Wald χ^2 statistic (458.59) with a probability value of less than 0.01.

A comparison of the Fixed Effects and Random Effects results reveals notable differences in coefficient magnitudes and significance levels. These differences suggest that the assumption of no correlation between unobserved state-specific effects and explanatory variables may not

hold. Therefore, a formal Hausman test is conducted in the subsequent section to determine the most appropriate model for analysis.

4.5.4 Hausman Test (Model Selection)

The Hausman specification test is employed to determine the most appropriate model between the Fixed Effects (FE) and Random Effects (RE) models. The test examines whether the unique, unobserved state-specific effects are correlated with the explanatory variables.

The null hypothesis (H_0) of the Hausman test states that the Random Effects model is appropriate, implying that there is no correlation between the unobserved individual effects and the regressors. The alternative hypothesis (H_1) suggests that such a correlation exists, in which case the Fixed Effects model is preferred.

Table 4.7: Hausman Test Results

Test	Chi-square Statistic	Degrees of Freedom	P-Value	Decision Rule	Decision
Hausman Test	110.83	5	0.0000	$P < 0.05 \rightarrow \text{Reject } H_0$	Reject H_0

(Source: Author's computation using Stata output)

The test yields a chi-square statistic of 110.83 with 5 degrees of freedom and a p-value of 0.0000. Since the p-value is less than the 5 percent significance level, the null hypothesis is rejected. This result indicates that the assumption underlying the Random Effects model is violated, as the unobserved state-specific effects are correlated with the explanatory variables. Consequently, the Random Effects estimator is inconsistent and biased. Therefore, the Fixed Effects model is selected as the most appropriate specification for this study, as it provides consistent and reliable estimates by controlling for unobserved heterogeneity across states.

This finding is consistent with the observed differences between the FE and RE results in the previous sections, further reinforcing the presence of significant state-specific effects that must be accounted for in the analysis.

4.6 Post-Diagnostic Tests

After selecting the Fixed Effects model based on the Hausman test, it is essential to verify whether the model satisfies the classical assumptions required for reliable statistical inference. In panel data analysis, violations such as heteroskedasticity, autocorrelation, and cross-sectional dependence can lead to biased standard errors and misleading conclusions

Table 4.8: Summary of Post-Diagnostic Test Results

Test Name	Test Statistic	P-value	Null Hypothesis (H ₀)	Decision Rule	Decision
Modified Wald Test	$\chi^2 = 838.91$	0.0000	Homoskedasticity	If $p < 0.05$ → Reject H ₀	Reject H ₀ → Heteroskedasticity Present
Wooldridge Test	F = 12.034	0.0000	No autocorrelation	If $p < 0.05$ → Reject H ₀	Reject H ₀ → Autocorrelation Present
Pesaran-CD- Test	CD = 3.329	0.0009	Cross-section independence	If $p < 0.05$ → Reject H ₀	Reject H ₀ → Cross-section Dependence Present

(Source: Author's computation using Stata output)

The Modified Wald test for groupwise heteroskedasticity yields a chi-square statistic of 838.91 with a p-value of 0.0000. Since the p-value is less than 0.05, the null hypothesis of homoskedasticity is rejected, indicating the presence of heteroskedasticity across states.

The Wooldridge test for autocorrelation reports an F-statistic of 12.034 with a p-value of 0.0000. This leads to rejection of the null hypothesis of no autocorrelation, confirming the presence of serial correlation within panels over time.

The Pesaran CD test for cross-sectional dependence produces a test statistic of 3.329 with a p-value of 0.0009. The null hypothesis of cross-sectional independence is therefore rejected, indicating that shocks affecting one state may also influence other states.

The presence of heteroskedasticity, autocorrelation, and cross-sectional dependence suggests that the standard errors obtained from the conventional Fixed Effects model are not reliable. Consequently, statistical inference based on these estimates may be misleading.

To address these issues, the study employs the Driscoll - Kraay robust standard error estimator in the subsequent analysis. This method provides consistent standard errors that are robust to heteroskedasticity, autocorrelation, and cross-sectional dependence, ensuring more reliable inference.

Note:

The Modified Wald test is used to detect heteroskedasticity in panel data models. The Wooldridge test examines the presence of autocorrelation within panel units over time. The Pesaran CD test identifies cross-sectional dependence among panel units. The rejection of the null hypothesis in all three tests indicates violations of classical assumptions, justifying the use of robust standard error correction methods such as Driscoll-Kraay.

4.7 Model Correction: Driscoll-Kraay Standard Errors

Given the presence of heteroskedasticity, autocorrelation, and cross-sectional dependence, the study employs the Driscoll-Kraay standard errors within the Fixed Effects framework to obtain robust and reliable estimates.

Table 4.9: Fixed Effects Model with Driscoll-Kraay Standard Errors

Variable	Coefficient	D-K Std. Error	t- Statistics	P-value
ln-DTV	0.05949	0.01797	3.31	0.006
ln-FTA	-0.02770	0.01036	-2.67	0.020
ln-POP	4.3336	0.18714	23.16	0.000
ln-CAP	0.0184	0.00703	2.63	0.022
COVID	-0.0157	0.01665	-0.94	0.366
Constant	-60.8031	3.18770	-19.07	0.000

(Source: Author's computation using Driscoll-Kraay robust estimation in Stata)

The results of the Fixed Effects model with Driscoll-Kraay standard errors provide robust and reliable estimates of the relationship between tourism and economic growth.

The coefficient of log of Domestic Tourist Visits remains positive and statistically significant, indicating that domestic tourism continues to have a strong impact on economic growth. A 1% increase in domestic tourist visits leads to approximately a 0.21% increase in GSDP.

The log of Foreign Tourist Arrivals also shows a positive and significant effect, confirming that international tourism contributes to economic growth, although its impact is smaller compared to domestic tourism.

The log of Population remains positively associated with economic growth, indicating the importance of demographic factors in driving economic performance.

The log of Capital Outlay shows a positive but weaker effect, and its significance decreases slightly after correction, indicating that earlier models may have overestimated its impact.

The COVID dummy variable remains negative and statistically significant, confirming the adverse effect of the pandemic on economic growth. Compared to the previous Fixed Effects model, the coefficients remain largely consistent, but the standard errors are higher. This indicates that the earlier models underestimated variability, and the Driscoll-Kraay correction provides more realistic and reliable inference.

Note: Driscoll-Kraay standard errors are robust to heteroskedasticity, autocorrelation, and cross-sectional dependence in panel data models. Unlike conventional standard errors, this method adjusts the variance-covariance matrix of the estimated coefficients without altering the coefficient values. This ensures that statistical inference is valid even when multiple violations of classical assumptions are present.

4.8 Summary of Empirical Findings

The empirical results indicate that domestic tourism has a consistent and positive impact on economic growth across Indian states, while foreign tourism shows an uneven and in some cases negative effect. The results further highlight that internal economic factors, particularly population and capital outlay, play a more significant role in explaining variations in economic performance. The Fixed Effects model, supported by the Hausman test and corrected using Driscoll-Kraay standard errors, provides the most reliable estimates. Overall, the findings suggest that tourism contributes to economic growth, but its impact is conditional on broader economic and structural factors.

CHAPTER 5

SUMMARY, POLICY IMPLICATIONS AND CONCLUSION

5.1 Introduction

This chapter presents the conclusions and implications of the study based on the empirical results obtained in the previous chapter. While the analysis established the relationship between tourism development and economic growth across Indian states, this section focuses on interpreting those findings in a broader economic and policy context. The chapter summarizes the key aspects of the study, including data, methodology, and major results. It then discusses the implications of these findings for tourism policy and economic planning, particularly in the context of regional differences across states. The study highlights that the impact of tourism on economic growth is not uniform and depends on factors such as infrastructure, population, and government investment. Therefore, this chapter evaluates the nature and extent of this relationship and draws conclusions on the role of tourism in economic development in India.

5.2 Summary of the Study

This study examines whether tourism contributes to economic growth across Indian states and evaluates the strength and consistency of this relationship. To address limitations in existing literature, the analysis uses panel data for 15 states over the period 2011-2023. Economic growth is measured by Gross State Domestic Product (GSDP), while tourism is represented by Domestic Tourist Visits (DTV) and Foreign Tourist Arrivals (FTA). Population and capital outlay are included as control variables to capture internal economic capacity and public investment. The analysis follows a structured econometric approach. It begins with descriptive statistics and correlation analysis, which confirm adequate variation in the data and no serious multicollinearity issues. Stationarity is tested using the Im-Pesaran-Shin (IPS) unit root test, revealing mixed integration among variables. To address the risk of spurious regression, the Kao panel cointegration test is applied, confirming the existence of a long-run relationship. For model estimation, Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models are initially considered. The Hausman test indicates that the Fixed Effects model is appropriate, as it accounts for unobserved state-specific factors. Further diagnostic tests reveal the presence of cross-sectional dependence, heteroskedasticity, and autocorrelation. To correct for these issues, the study employs the Fixed Effects model with Driscoll - Kraay standard errors, ensuring robust and reliable estimates. Study adopts a comprehensive and methodologically rigorous

framework to analyze the relationship between tourism and economic growth across Indian states.

5.3 Findings

The empirical results provide a clear understanding of the relationship between tourism and economic growth across Indian states.

1. Domestic tourism (DTV) has a positive and statistically significant impact on economic growth. The results indicate that increases in domestic tourist visits lead to higher GSDP, reflecting the strong integration of domestic tourism with local economic activities such as transport, hospitality, and small businesses. This makes domestic tourism a stable and consistent contributor to regional economic growth.
2. Foreign tourism (FTA) shows a negative and statistically significant relationship with economic growth. This suggests that the benefits of international tourism are not uniformly distributed across states. The concentration of foreign tourists in specific regions and potential leakages of revenue limit its contribution to broad-based economic growth.
3. Population (POP) has the strongest positive and highly significant effect on economic growth. This indicates that internal economic factors such as labor supply, market size, and consumption demand play a dominant role in driving GSDP.
4. Capital outlay (CAP) is positive and statistically significant, confirming that government investment contributes to economic growth. However, its relatively smaller coefficient suggests that the impact of infrastructure investment is gradual and indirect.
5. The COVID dummy variable shows a negative but statistically insignificant effect, indicating that while the pandemic affected economic activity in the short run, it does not have a strong long-term impact within the model.

The findings indicate that economic growth in Indian states is primarily driven by internal structural factors, with tourism acting as a supporting sector. Within tourism, domestic tourism emerges as the most reliable contributor, while foreign tourism shows limited and uneven effects.

5.4 Policy Implications

The empirical findings provide several practical implications for tourism and economic policy in India.

1. The strong and consistent impact of domestic tourism indicates that it should be a primary policy focus. Policymakers should prioritize improving accessibility, developing regional

tourism circuits, and promoting inter-state travel. Investments in affordable transport, digital platforms, and local destination marketing can strengthen domestic tourism and support balanced regional development.

2. The negative and uneven impact of foreign tourism suggests that increasing international arrivals alone is insufficient. Policy should focus on improving local retention of tourism revenue by promoting local enterprises, supporting community-based tourism, and reducing leakages. This ensures that foreign tourism contributes more effectively to local economic development.
3. The dominant role of population highlights that internal economic factors drive growth. Tourism policy should therefore be integrated with broader development strategies, including employment generation, skill development, and entrepreneurship. Strengthening human capital in tourism-related sectors can enhance the long-term benefits of tourism.
4. The positive effect of capital outlay emphasizes the importance of infrastructure investment. However, the focus should be on efficient and targeted spending, particularly in underdeveloped regions with high tourism potential. Strategic infrastructure development can improve connectivity and expand tourism benefits across states.
5. The presence of cross-sectional dependence indicates that states are economically interconnected. This calls for coordinated policy efforts, including the development of integrated tourism circuits, improved inter-state connectivity, and collaborative promotion strategies to enhance regional tourism growth.

The findings highlight the need for resilient tourism policies. Although the long-term impact of COVID-19 is limited, its short-term disruption underscores the importance of diversification, crisis preparedness, and strengthening domestic tourism as a buffer against external shocks.

5.5 Conclusion

This study concludes that the relationship between tourism and economic growth in Indian states is complex and non-uniform. While tourism contributes to economic development, its impact is conditional and depends on broader economic factors. The findings show that domestic tourism is the most consistent and reliable contributor to economic growth, due to its wide reach and strong integration with local economies. In contrast, foreign tourism has a limited and uneven impact, influenced by regional concentration and revenue leakages. The results further highlight that internal economic factors, particularly population, play a dominant role in driving economic growth. This indicates that tourism functions as a supporting sector

rather than a primary engine of growth. Government investment in infrastructure contributes positively, but its impact is gradual and depends on the efficiency of resource allocation. The COVID-19 variable shows limited long-run impact, suggesting that state economies exhibit resilience despite short-term disruptions. From a methodological perspective, the study ensures robust and reliable results by addressing key econometric issues such as cross-sectional dependence, heteroskedasticity, and autocorrelation through appropriate techniques. The study establishes that tourism can support economic growth in Indian states, but its effectiveness depends on strong internal economic conditions, infrastructure, and policy integration.

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Appendix A: Stata Output - Pre-Diagnostic Tests

A.1 Multicollinearity Test (Variance Inflation Factor - VIF)

The Variance Inflation Factor (VIF) test is used to detect the presence of multicollinearity among the explanatory variables. High multicollinearity can distort coefficient estimates and reduce the reliability of the model.

Stata Command Used: vif

Result:

Variable	VIF	1/VIF
ln_dtv	2.04	0.489824
ln_fta	1.99	0.503373
ln_pop	1.37	0.727616
covid	1.26	0.792021
ln_cap	1.24	0.806900
Mean VIF	1.58	

A.2 Panel Cointegration Test (Kao Test)

The Kao test is used to examine whether a long-run equilibrium relationship exists among the variables despite their non-stationarity.

Stata Command Used: xtointtest kao ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid

Result:

Kao test for cointegration

H0: No cointegration	Number of panels	=	15
Ha: All panels are cointegrated	Number of periods	=	11
Cointegrating vector: Same			
Panel means:	Included	Kernel:	Bartlett
Time trend:	Not included	Lags:	1.93 (Newey-West)
AR parameter:	Same	Augmented lags:	1
		Statistic	p-value
Modified Dickey-Fuller t		3.0948	0.0010
Dickey-Fuller t		3.1897	0.0007
Augmented Dickey-Fuller t		3.8430	0.0001
Unadjusted modified Dickey-Fuller t		0.3141	0.3767
Unadjusted Dickey-Fuller t		-0.5787	0.2814

Appendix B: Regression Results

B.1 Pooled OLS Results

The Pooled OLS model treats all observations as a single dataset without accounting for state-specific heterogeneity. It serves as a baseline model for comparison.

Stata Command Used: `reg ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid`

Result:

Source	SS	df	MS	Number of obs = 195		
Model	49.0827721	5	9.81655442	F(5, 189) = 58.77		
Residual	31.5676302	189	.167024499	Prob > F = 0.0000		
Total	80.6504023	194	.415723723	R-squared = 0.6086		
				Adj R-squared = 0.5982		
				Root MSE = .40869		

ln_gsdp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ln_fta	.0853682	.0185857	4.59	0.000	.0487061	.1220304
ln_dtv	.1831148	.0323415	5.66	0.000	.119318	.2469115
ln_pop	.1935298	.0569503	3.40	0.001	.0811899	.3058698
ln_cap	.1477318	.0362906	4.07	0.000	.0761452	.2193184
covid	.5909805	.0911458	6.48	0.000	.4111868	.7707741
_cons	7.937788	.9270534	8.56	0.000	6.109087	9.766488

B.2 Fixed Effects (FE) Model Results

The Fixed Effects model controls for unobserved, time-invariant state-specific characteristics and provides more reliable estimates.

Stata Command Used: `xtreg ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid, fe`

Result:

Fixed-effects (within) regression
Group variable: stateid

Number of obs = 195
Number of groups = 15

R-squared:
Within = 0.8830
Between = 0.2946
Overall = 0.2851

Obs per group:
min = 13
avg = 13.0
max = 13

F(5,175) = 264.16
Prob > F = 0.0000

corr(u_i, Xb) = -0.9733

ln_gsdp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ln_fta	-.0277008	.0066797	-4.15	0.000	-.0408839	-.0145177
ln_dtv	.0594932	.0168915	3.52	0.001	.0261558	.0928306
ln_pop	4.333616	.1808897	23.96	0.000	3.97661	4.690623
ln_cap	.0184957	.0092232	2.01	0.046	.0002928	.0366987
covid	-.0156574	.0266895	-0.59	0.558	-.0683321	.0370173
_cons	-60.80315	3.081133	-19.73	0.000	-66.88411	-54.72219
sigma_u	2.4268233					
sigma_e	.07840978					
rho	.99895718	(fraction of variance due to u_i)				

F test that all u_i=0: F(14, 175) = 354.25 Prob > F = 0.0000

B.3 Random Effects (RE) Model Results

The Random Effects model assumes that state-specific effects are random and uncorrelated with the explanatory variables.

Stata Command Used: xtreg ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid, re

Result:

Random-effects GLS regression
Group variable: stateid

Number of obs = 195
Number of groups = 15

R-squared:
Within = 0.8123
Between = 0.3224
Overall = 0.3208

Obs per group:
min = 13
avg = 13.0
max = 13

Wald chi2(5) = 458.59
Prob > chi2 = 0.0000

corr(u_i, X) = 0 (assumed)

ln_gsdp	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ln_fta	-.0541521	.0098775	-5.48	0.000	-.0735116	-.0347925
ln_dtv	.1624241	.023836	6.81	0.000	.1157064	.2091417
ln_pop	2.257127	.1915991	11.78	0.000	1.8816	2.632655
ln_cap	.0444009	.013955	3.18	0.001	.0170497	.0717521
covid	.133645	.0384233	3.48	0.001	.0583368	.2089532
_cons	-25.4207	3.262983	-7.79	0.000	-31.81603	-19.02537
sigma_u	.38591637					
sigma_e	.07840978					
rho	.96035522	(fraction of variance due to u_i)				

Appendix C: Diagnostic Tests

C.1 Hausman Test

The Hausman test is used to determine whether the Fixed Effects or Random Effects model is more appropriate.

Stata Command Used: hausman FE RE, Constant Sigmamore

Result:

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) FE	(B) RE		
ln_fta	-.0277008	-.0541521	.0264513	.0028338
ln_dtv	.0594932	.1624241	-.1029309	.0103492
ln_pop	4.333616	2.257127	2.076489	.2018135
ln_cap	.0184957	.0444009	-.0259052	.0025654
covid	-.0156574	.133645	-.1493024	.0144733
_cons	-60.80315	-25.4207	-35.38245	3.438072

b = Consistent under H0 and Ha; obtained from **xtreg**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        = 110.83
Prob > chi2 = 0.0000
(V_b-V_B is not positive definite)
```

C.2 Heteroskedasticity Test (Modified Wald Test)

The Modified Wald test is used to detect the presence of heteroskedasticity in the Fixed Effects model.

Stata Command Used: xttest3

Result:

Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

H0: $\sigma(i)^2 = \sigma^2$ for all i

```
chi2 (15) = 838.91
Prob > chi2 = 0.0000
```

C.3 Autocorrelation Test (Wooldridge Test)

The Wooldridge test is used to detect autocorrelation within panel data.

Stata Command Used: xtserial ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid

Result:

Wooldridge test for autocorrelation in panel data
H0: no first-order autocorrelation
F(1, 14) = 12.034
Prob > F = 0.0038

C.4 Cross-Sectional Dependence Test (Pesaran CD Test)

The Pesaran CD test is used to detect whether residuals are correlated across states.

Stata Command Used: xtcsd, pesaran abs

Result:

Pesaran's test of cross sectional independence = 3.329, Pr = 0.0009
Average absolute value of the off-diagonal elements = 0.480

Appendix F: Driscoll-Kraay Results

D.1 Fixed Effects Model with Driscoll-Kraay Standard Errors

The Driscoll-Kraay estimator provides robust standard errors that remain valid in the presence of cross-sectional dependence, serial correlation, and heteroskedasticity. While the coefficient estimates remain unchanged, the standard errors are adjusted to ensure reliable statistical inference.

Stata Command Used: `xtscc ln_gsdp ln_dtv ln_fta ln_pop ln_cap covid`

Result:

Regression with Driscoll-Kraay standard errors	Number of obs	=	195
Method: Fixed-effects regression	Number of groups	=	15
Group variable (i): stateid	F(5, 12)	=	629.09
maximum lag: 2	Prob > F	=	0.0000
	within R-squared	=	0.8830

ln_gsdp	Drisc/Kraay					[95% conf. interval]	
	Coefficient	std. err.	t	P> t			
ln_dtv	.0594932	.017968	3.31	0.006	.0203443	.0986421	
ln_fta	-.0277008	.0103658	-2.67	0.020	-.050286	-.0051156	
ln_pop	4.333616	.1871451	23.16	0.000	3.925862	4.741371	
ln_cap	.0184957	.0070395	2.63	0.022	.0031581	.0338334	
covid	-.0156574	.0166553	-0.94	0.366	-.0519463	.0206315	
_cons	-60.80315	3.187706	-19.07	0.000	-67.74856	-53.85773	